

SUMMARY

I am a last year PhD candidate in the MAMBO team at Toulouse University (expected graduation: Dec. 2026), advised by [Prof. Pierre Weiss](#) and [Prof. Edouard Pauwels](#). My research focuses on computational imaging, mathematical analysis, and modern generative modeling (including score-based diffusion models and flow matching). I specialize in developing theoretical frameworks and deep learning models for inverse problems in imaging, with applications in microscopy and satellite imaging. I'm also interested in generative modeling for image generation, image reconstruction and vision-language models (VLMs).

EDUCATION

2023 - present Toulouse University, France
PhD in Applied Mathematics, advised by [Prof. Pierre Weiss](#) and [Prof. Edouard Pauwels](#).
2018 - 2023 INSA Toulouse, France
Master of Engineering in Applied Mathematics (incl. bachelor), top 3% of the year

WORK EXPERIENCE

Teaching assistant – INSA Toulouse Dec 2023 - May 2026
Courses: Signal Processing, Stochastic Optimization and Machine Learning

Research Engineer – Agenium Space Dec 2022 - Sept 2023
Correcting micro-vibrations in satellite imaging with pushbroom cameras. Modeled the physical imaging process, derived theoretical bounds for image restoration, and designed diffusion-based generative methods to improve real data from Sentinel-2B.

Research Intern – Toulouse Mathematics Institute Jun 2022 - Sept 2022
Product-convolution neural networks: applications to image deblurring with space-varying blurs, with [Pierre Weiss](#) and [Paul Escande](#). Formulated local quantitative stability properties and designed a deep learning-based framework for deblurring with complex spatially varying blurs.

OPEN-SOURCE PROJECTS

Blind deblurring in Microscopy [Online demo](#)
Designing mathematical and deep learning methods to jointly estimate the sharp image and the unknown blur kernel. Focused on rigorous physical modeling of optical systems, PSF estimation and image deblurring. The framework supports various imaging modalities (RIM, TIRF, confocal) and space-varying blurs..

DeepInverse – Core maintainer [Link to Github](#)
An open-source library ([PyTorch Ecosystem](#)) for solving imaging inverse problems with state-of-the-art AI. Contributed advanced mathematical solvers, diffusion & flow matching models, and algorithmic designs ensuring research reproducibility and robust theoretical baselines.

SELECTED PUBLICATIONS

Nguyen, Minh Hai et al. (2026). *A Morse-Bott Framework for Blind Inverse Problems: Local Recovery Guarantees and the Failure of the MAP*. arXiv: [2508.02923 \[cs.CV\]](#). URL: <https://arxiv.org/abs/2508.02923>.

Nguyen, Minh Hai et al. (2026). “An analytic theory of convolutional neural network inverse problems solvers”. In: *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. arXiv: [2601.10334 \[cs.CV\]](#). URL: <https://arxiv.org/abs/2601.10334>.

Tachella, Julián et al. (2025). “DeepInverse: A Python package for solving imaging inverse problems with deep learning”. In: *Journal of Open Source Software* 10.115, p. 8923. DOI: [10.21105/joss.08923](https://doi.org/10.21105/joss.08923). URL: <https://doi.org/10.21105/joss.08923>.

Nguyen, Minh Hai and Pierre Weiss (2024). “Comparing Plug-and-Play and Unrolled networks”. In: *Pre-print*.

Nguyen, Minh Hai et al. (2024). “DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras”. In: *IEEE Transactions on Geoscience and Remote Sensing* 62, pp. 1–9.

SKILLS

Research: computational imaging, generative models (diffusions, flows, VLMs) and representation learning.
Programming language: Python, C++, SQL
Deep learning frameworks: PyTorch, JAX, CUDA/Triton kernels, HuggingFace ecosystem, distributed training (DDP, FSDP), profiling (Nsight, torch-profiler), numpy, scikit-learn, pandas, HPC clusters ([Jean-Zay](#)).
Tools: Linux, Git/GitHub, Docker, open-source ecosystem.
Languages: English (professional), French (fluent), Vietnamese (native).

TALKS AND PRESENTATIONS

- [ICML 2026 \(Poster\)](#) Seoul, South Korea, Jul. 2026
Title: An analytic theory of convolutional neural network inverse problems solvers
- [PyTorch Conference Europe 2026 \(Speaker\)](#) Paris, France, Apr. 2026
Title: DeepInverse and Computational Imaging
- Blind inverse problems in imaging: from foundations to applications (Oral) Marseille, France, Oct 2025
Title: How diffusion prior landscapes shape the posterior in blind deconvolution

ACADEMIC SERVICE

Reviewing

- Conferences: ICML 2026, CVPR 2025, GRETSI 2025
- Journals: Journal of Mathematical Imaging and Vision (JMIV)

DISTINCTIONS

- French National Prize for Open Science Library for DeepInverse Nov 2024
- Doctoral Research Fellowship, French Ministry of Higher Education and Research 2023 - 2026
- IA Pau Winner: efficient language model (50+ teams) 2023